

SANYAM KAUL

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EDUCATION

M.Tech Computer Science & Engineering *IIT Hyderabad*
2025 CGPA: 8.50 **B.Tech Computer Science & Engineering (Networks & Security)** *VIT-AP*
University, Amaravati
2021 CGPA: 8.90

EXPERIENCE

Data Engineer *Netradyne*
Jul 2025 – Present

- Enhanced alert-kpi tool with _ALL locale support for dataset creation, enabling model training teams to train on aggregated datasets
- Optimized TensorRT-based reprocessing pipelines (v7, v8, v10) for analytics code compilation and Docker image management
- Built AWS Monitoring Dashboard with cost analysis and machine insights for data infrastructure optimization

Tech Analyst (SDE-1) *BlackRock*
Jul 2021 – Jun 2023

- Engineered Python-based applications with 85%+ test coverage, focusing on scalable architecture and design patterns
- Implemented automated CI/CD pipelines achieving 50% automation in performance and integration testing

PROJECTS

WineSecrethK - RAG-based AI Food & Wine Pairing System *rag.winesecrethk.com*

- Built RAG-based AI system with HyDE-inspired algorithm for flavor profile matching using cosine similarity on vector embeddings
- Implemented multi-modal input processing (text, audio, images) with automatic query intent identification
- Algorithm submitted for patent; Tech stack: Python, FastAPI, AWS RDS, ChromaDB vector database

IITH FoML Hackathon 2023 - 1st Place *IIT Hyderabad*
2023

- Won 1st place with voting-based ensemble model combining Random Forest and Gradient Boosting classifiers
- Implemented advanced ensemble learning techniques to improve prediction accuracy through model combination

IITH Deep Learning Hackathon 2024 - 15th/100+ Teams *IIT Hyderabad*
2024

- Ranked 15th out of 100+ teams by implementing and comparing ResNet variants, DenseNet, ResNext, and Vision Transformer (ViT)
- Performed extensive hyperparameter tuning and model comparison to achieve competitive results on image classification tasks

Deep Learning Architectures from Scratch *IIT Hyderabad*
2023-2024

- Engineered complex architectures from scratch: CNN, RNN, Vision Transformer, Autoencoder, Variational Autoencoder
- Implemented core algorithms including backpropagation, MLP, and optimization techniques using NumPy
- Gained deep understanding of neural network internals, gradient computation, and comparative model analysis

Poisson Regression for Meal Time Prediction *IIT Hyderabad*
2023

- Developed Poisson Regression model for meal time prediction with Gradient Descent implemented from scratch
- Demonstrated strong mathematical foundations by building ML algorithms without external libraries

SVHN Dataset Analysis *IIT Hyderabad*
2023

- Analyzed Street View House Numbers dataset using PCA for dimensionality reduction and feature extraction
- Implemented K-NN and K-means clustering for pattern recognition and unsupervised learning

TECHNICAL SKILLS

- ML/DL Frameworks:** PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas
- ML Algorithms:** Random Forest, Gradient Boosting, Ensemble Methods, CNN, RNN, ViT, Autoencoders, PCA, K-means, K-NN
- AI/ML Concepts:** RAG, Vector Embeddings, Backpropagation, Gradient Descent, Hyperparameter Tuning, Model Evaluation

- **Languages:** Python (Primary), Java, TypeScript, JavaScript
- **Tools:** AWS (S3, RDS), ChromaDB, FastAPI, Git, Docker, Jupyter Notebooks

ACHIEVEMENTS

- 1st position in IITH ML Hackathon 2023; 15th position (out of 100+ teams) in IITH Deep Learning Hackathon 2024
- 1st position in Object Oriented Programming course (100+ students); 450+ problems solved on LeetCode
- 4/5 star rating in BlackRock annual employee ratings (2021, 2022)